

M52: Final Composite Dependent Variable

PCA on Raw Switch Counts (Time, Topic-LDA, Typing)

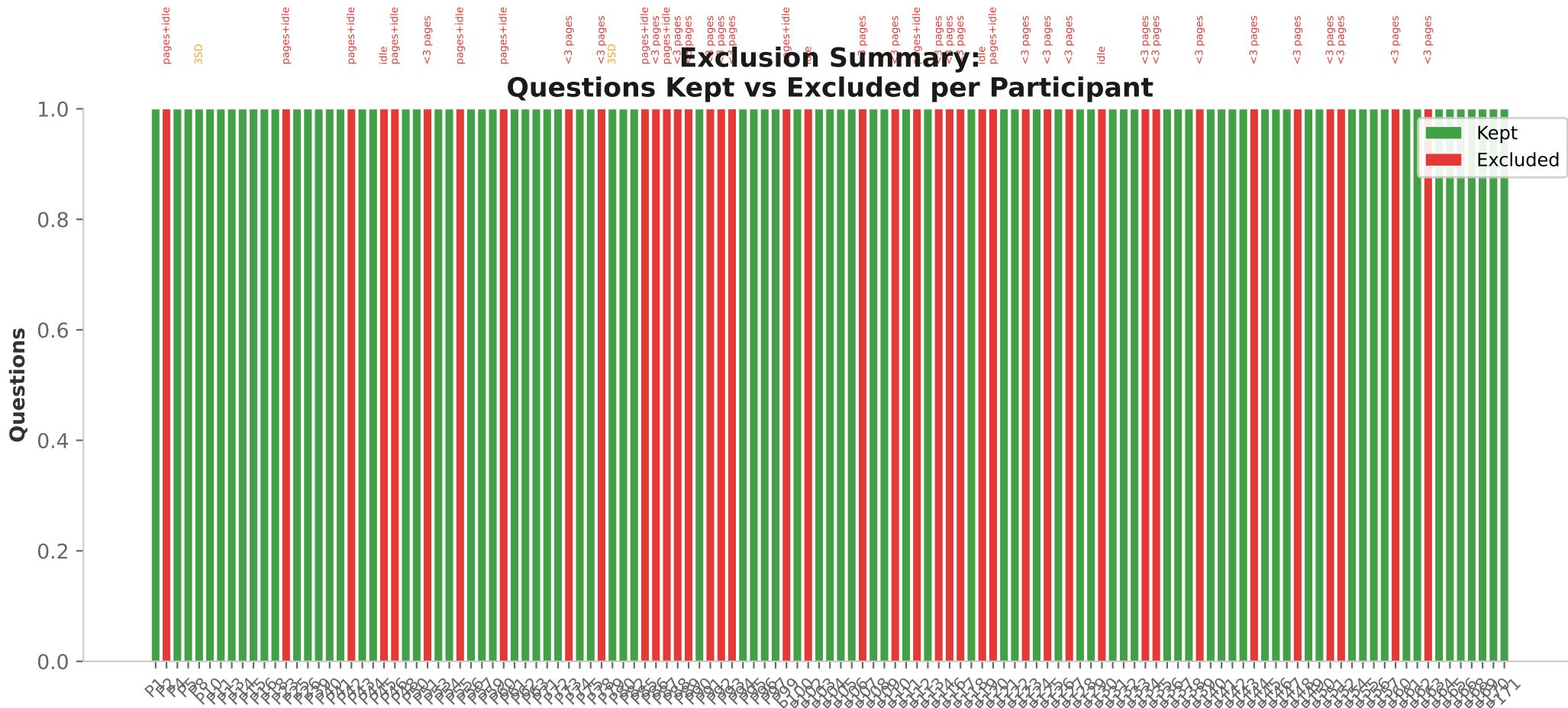
No standardization (z-score) applied

Exclusion Criteria Applied:

1. Questions with fewer than 3 page visits: 37 excluded
2. Questions with idle time $\geq 50\%$ after last meaningful event: 15 excluded
3. Participants > 3 SD from mean on any DV: 2 excluded

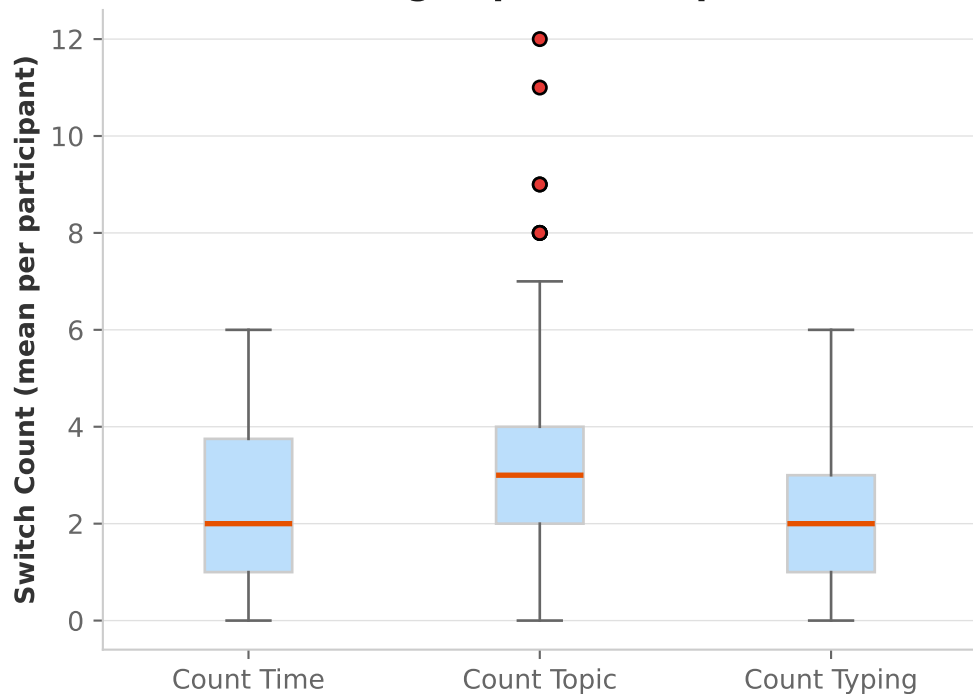
Result: 84 / 125 questions retained

Final sample: N = 82 participants

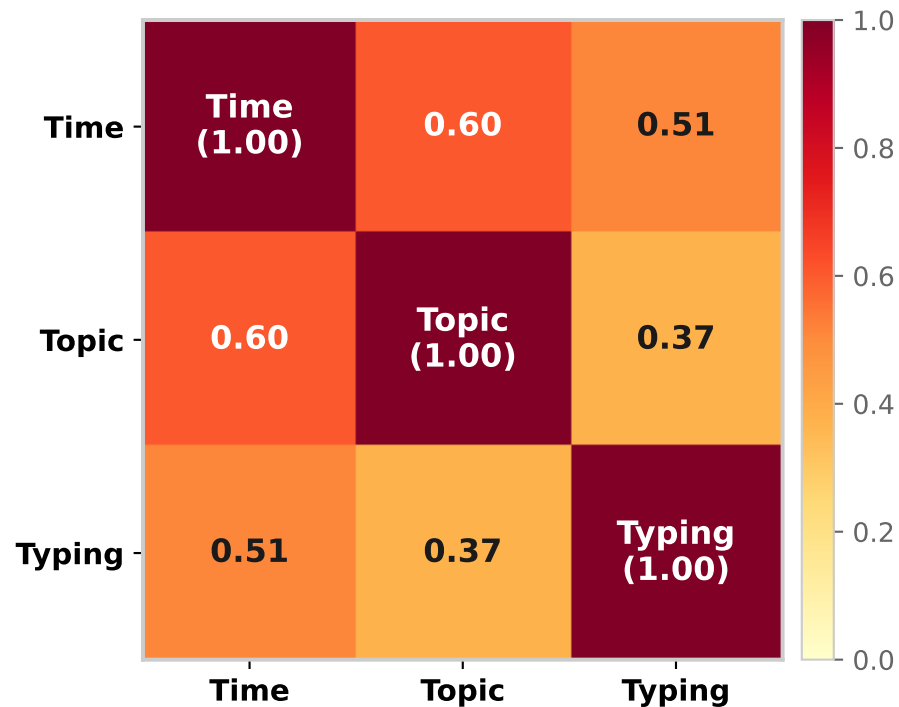


Descriptive Statistics: Raw Switch Counts

Distribution of Raw Switch Counts (Averaged per Participant)

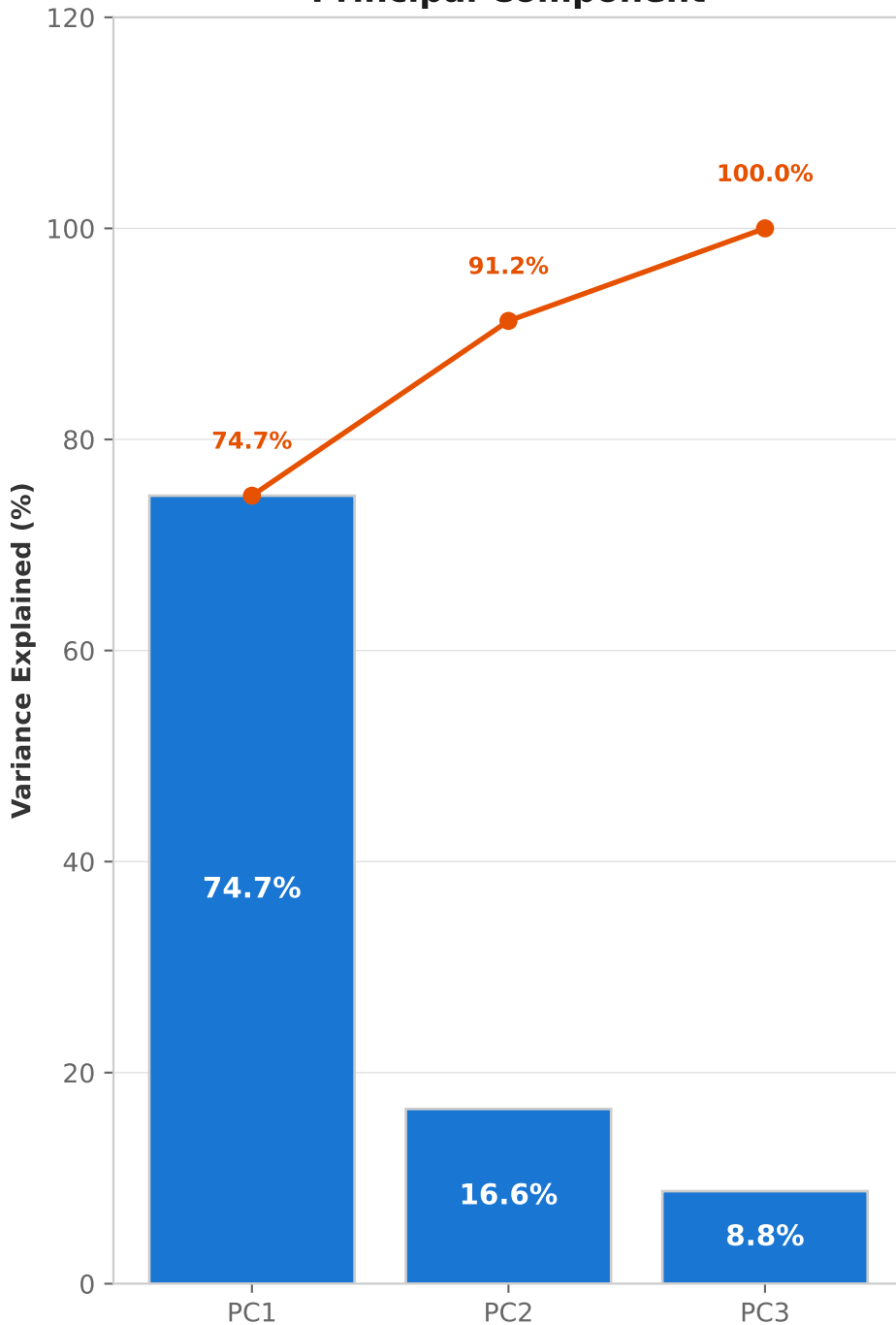


Correlation Between Switch Count Variables



PCA Results: Dimensionality Reduction on Raw Switch Counts

Variance Explained by Each Principal Component



PCA Loadings (variable weights per component):

PC1 (74.7% variance):

Count Topic	+0.864	+++++
Count Time	+0.438	+++++
Count Typing	+0.248	++++

PC2 (16.6% variance):

Count Typing	+0.664	+++++
Count Time	+0.573	+++++
Count Topic	-0.481	-----

PC3 (8.8% variance):

Count Typing	+0.706	+++++
Count Time	-0.693	-----
Count Topic	+0.148	++

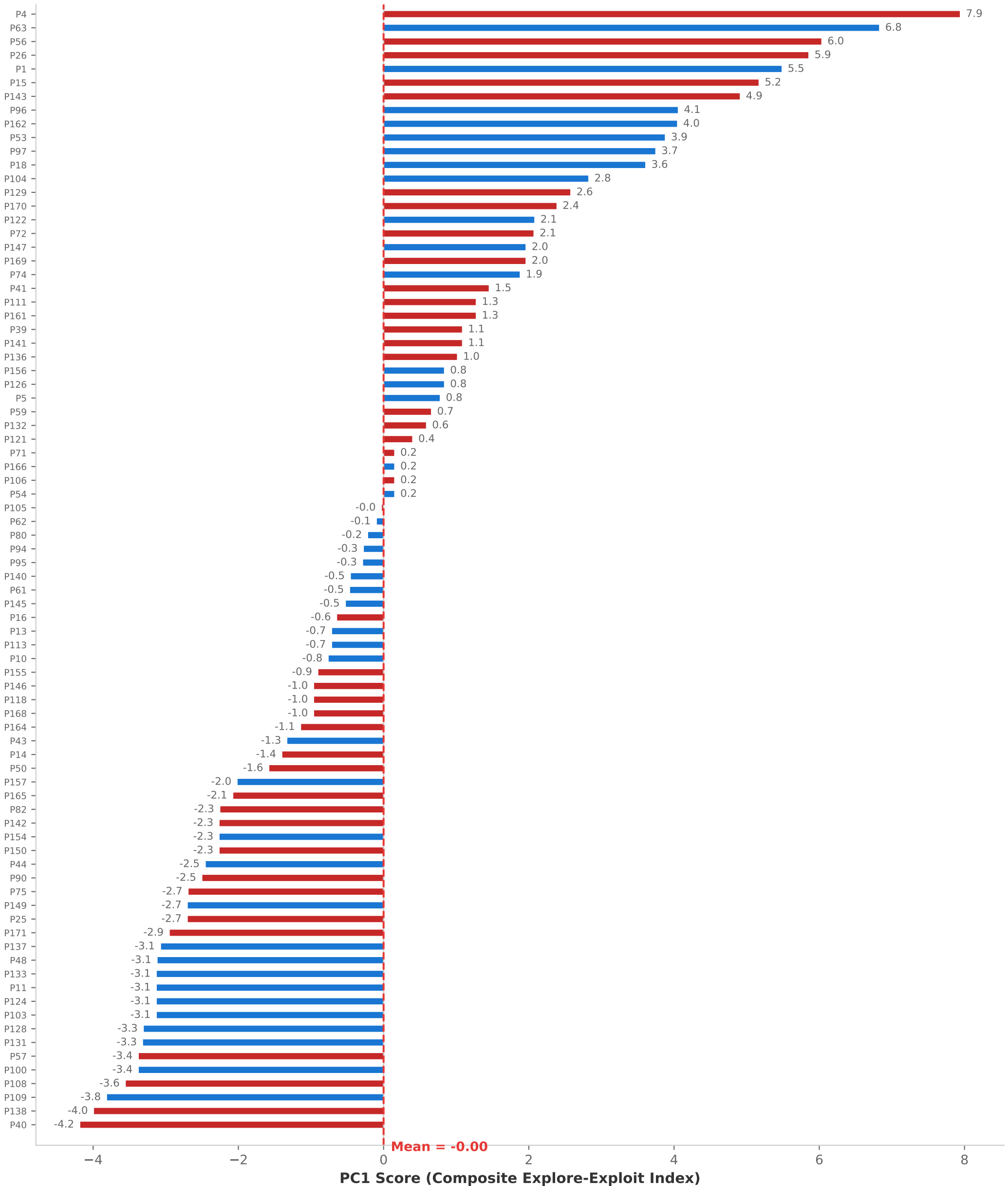
Interpretation:

PC1: Primarily reflects topic switching
(loading = +0.864)
Higher score = more topic transitions

PC2: Primarily reflects typing switching
(loading = +0.664)
Higher score = more typing transitions

PC3: Primarily reflects typing switching
(loading = +0.706)
Higher score = more typing transitions

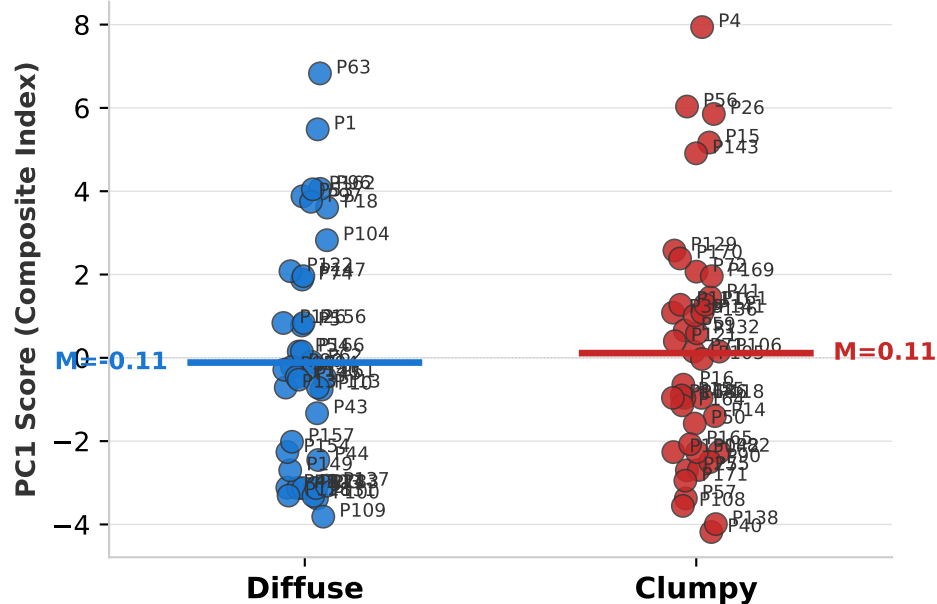
Individual PC1 Scores: Composite Explore-Exploit Index



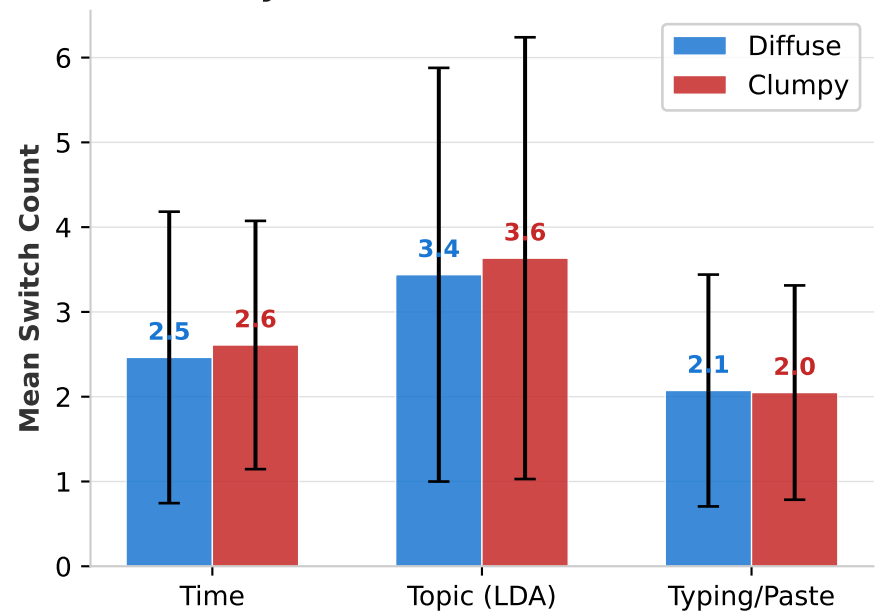
Condition Comparison: Clumpy vs Diffuse

N = 82 (Diffuse: 41, Clumpy: 41)

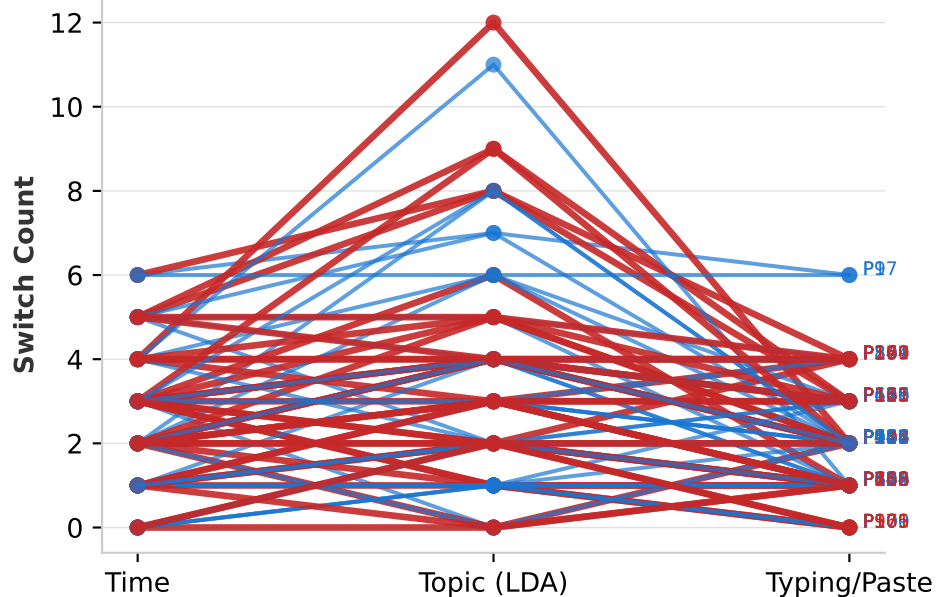
PC1: Composite Explore-Exploit
by Condition



Raw Switch Counts
by Condition (Mean +/- SD)



Individual Profiles
by Switch Type



Conditions

- Diffuse (n=41)
- Clumpy (n=41)

Condition Comparison: Descriptive Statistics & t-Tests

N = 82 (Diffuse: 41, Clumpy: 41)

DESCRIPTIVE STATISTICS

Variable	Diffuse (n=41)	Clumpy (n=41)
Time	M = 2.46, SD = 1.72	M = 2.61, SD = 1.46
Topic (LDA)	M = 3.44, SD = 2.44	M = 3.63, SD = 2.61
Typing/Paste	M = 2.07, SD = 1.37	M = 2.05, SD = 1.26
PC1 (composite)	M = -0.11, SD = 2.76	M = 0.11, SD = 2.87

INDEPENDENT-SAMPLES t-TESTS (Welch)

Variable	t	df	p	Cohen d	Sig.
Time	-0.41	78.0	0.6793	+0.09	n.s.
Topic (LDA)	-0.35	79.7	0.7272	+0.08	n.s.
Typing/Paste	0.08	79.5	0.9334	-0.02	n.s.
PC1 (composite)	-0.36	79.9	0.7169	+0.08	n.s.

* p < .05 ** p < .01 *** p < .001

INTERPRETATION

PC1 difference (Clumpy - Diffuse): +0.23
Clumpy condition shows more explore-exploit switches
Effect size (Cohen's d): +0.08
Result is NOT statistically significant (p = 0.7169)
Low statistical power due to small sample (N = 82)

Largest sub-variable effect: Time
d = +0.09, p = 0.6793

Note: Typing metric uses answer_snapshot bursts + paste events.